

CLIMACTIX GLOBAL

CLIMATE INTELLIGENCE ANALYSIS REPORT

BHARAT STEEL & ALLOYS LTD.

BSE / NSE: BSAL · Steel – Integrated Producer (GICS: 15101030) · India

CLIMATE INTELLIGENCE SCORE

B

40.3 / 100

CONFIDENCE

HIGH

Score: 78 / 100

PEER RANK	CONFIDENCE	ASSESSMENT YEAR	OUTLOOK
28th Percentile	78 / 100	FY2024	STABLE / POS.
Steel-IN-Integrated-Large	Data Freshness: Excellent	Apr 2023 – Mar 2024	Positive potential

REPORT DATE	June 2026	ENGINE VERSION	Climactix Intelligence Engine v1.0
ASSESSMENT ID	CX-2026-BSAL-001	METHODOLOGY	ARCHITECTURE.md v1.0
REPORT TYPE	Full-Spectrum Climate Intelligence	NEXT REVIEW	June 2027 / Milestone Trigger
PREPARED BY	Climactix Global Intelligence Engine	CLASSIFICATION	Institutional — Authorized Recipients Only

DISCLAIMER: This report is generated from structured assessment data submitted by Bharat Steel & Alloys Ltd. and publicly available climate science databases. All financial projections are model-based estimates carrying material uncertainty. This report does not constitute investment advice. Scores reflect climate risk management quality and disclosure — not creditworthiness or equity valuation. For authorized recipients only.

01 — EXECUTIVE SUMMARY

Bharat Steel & Alloys Ltd. has received a Climate Intelligence Score of B (40.3/100), placing it in the 28th percentile of its peer cohort (Steel — India — Integrated Large, n=11). Confidence in this rating is HIGH (78/100).

BSAL is a company in transition — one that discloses consistently, governs adequately, and has begun reducing emissions intensity — but has not yet built the institutional climate risk infrastructure required by the capital market standards now being set by SEBI, RBI, and global institutional investors. The company faces existential financial exposure under a Disorderly Transition scenario: combined physical and transition risk reaches **\$148M (23.9% of revenue) by 2030** and an unmanageable **\$911M (147% of revenue) by 2050** if no material decarbonization investment is made.

THREE STRUCTURAL VULNERABILITIES

01	CARBON LOCK-IN	BF-BOF technology transition
02	CLIMATE INTELLIGENCE DEFICIT	No scenario analysis, no interim milestones
03	SUPPLY CHAIN OPACITY	12% of Tier 1 suppliers, no supply chain risk assessment

FOR INSTITUTIONAL INVESTORS: BSAL is a borderline hold at current climate risk management maturity. The trajectory is positive but speed of progress is insufficient given the 2030 financial exposure horizon. Recommended: conditional engagement with 12-month reassessment trigger tied to SBTi submission, physical risk assessment completion, and third-party ESG assurance.

02 — CLIMATE INTELLIGENCE SCORECARD

OVERALL CIS	B	40.3 / 100	28th Percentile · DEVELOPING	HIGH (78)
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DOMAIN	RATING	SCORE	PEER PCT	MATURITY
Governance	BBB	59.8	55th	PROGRESSING
Physical Risk	BB+	49.5	47th	DEVELOPING
Transition Risk	CCC	28.2	16th	LAGGARD ■
Carbon Management	CCC	33.0	18th	LAGGARD ■
Supply Chain	D	19.3	12th	LAGGARD ■
Compliance	BB	50.2	48th	DEVELOPING
Adaptation	BB	46.9	52th	PROGRESSING
Disclosure Quality	A	74.4	78th	ADVANCED ✓
Greenwashing Risk	—	41.5	60th	ELEVATED ■
Climate Resilience	D	13.8	8th	LAGGARD ■

The B rating (40.3) reflects a company with functional governance and disclosure infrastructure but without climate risk management depth. The wide dispersion across domains is the defining characteristic — Disclosure Quality (A, 78th percentile) and Governance (BBB, 55th percentile) are institutional-grade. Carbon Management (CCC, 18th), Transition Risk (CCC, 16th), and Climate Resilience (D, 8th) are near-bottom of cohort — a classic profile of strong form, weak substance.

03 — GOVERNANCE

BBB	59.8 / 100	55th Percentile	PROGRESSING
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Strengths

BSAL's governance structure is substantively ahead of many Steel sector peers. The Environment and Sustainability Committee, established at Board level in FY2022, has verified Terms of Reference confirming mandate to review climate risk disclosures, approve sustainability targets, and monitor environmental compliance. The board-approved Sustainability Policy (FY2023) provides the foundational governance document from which disclosure obligations and target commitments are derived.

Material Gaps

Executive accountability is materially insufficient. The MD/CEO has 5% of variable compensation linked to PAT Scheme energy efficiency targets only. No other C-suite member has climate-linked KPIs. For a company with \$73M+ carbon pricing exposure by 2030, climate performance should constitute 15–25% of executive variable pay to be considered institutionally credible.

No third-party assurance exists on any sustainability data. BSAL's emissions, energy, and water data is entirely self-reported. There is no ISAE 3000 verification, no ISO 14064 audit, and no specialist firm review. This is the single most significant governance gap — and the highest-leverage action available: commissioning ISAE 3000 limited assurance would alone add 8–10 points to Governance and cascade to Confidence Score and Disclosure Quality.

Score Breakdown

COMPONENT	SCORE	WEIGHT	CONTRIBUTION
Board oversight	100.0	0.30	30.0
ESG policy quality	65.0	0.25	16.3
Executive accountability	40.0	0.20	8.0
Audit / verification	0.0	0.15	0.0
Stakeholder engagement	55.0	0.10	5.5

04 — CARBON MANAGEMENT

CCC	33.0 / 100	18th Percentile	LAGGARD
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Emissions Profile — FY2024

SCOPE	ABSOLUTE (TCO2E)	INTENSITY (TCO2E/T)	VS BENCHMARK
Scope 1 (Direct)	5,376,000	1.92	3.8% above median ■
Scope 2 Market-based	892,000	0.32	
Scope 2 Location-based	1,104,000	0.39	
Sector Median (p50)	—	1.85	Benchmark reference

BSAL's 3-year Scope 1 intensity decline of **2.94%/yr** is Paris-aligned (2–7%/yr range). This is the most positive data point in BSAL's carbon profile. However, the company remains above the sector median (1.92 vs 1.85 tCO2e/t), driving a **zero score on emissions performance** — the largest single sub-component (weight: 0.35). Crossing below 1.85 tCO2e/t (achievable on current trajectory by FY2027) would alone raise CMS from 33.0 to approximately 68 (BBB+).

Target Credibility Assessment

CRITERION	STATUS	SCORE
SBTi Validation	Not submitted	0
Interim 2030 milestone	Not defined	0
CAPEX allocation to decarbonization	Undisclosed	0
Technology transition roadmap	Not published	0
Board approval of target	Yes — policy uploaded	25

A Net-Zero 2050 commitment from a BF-BOF steel company without a technology transition plan, without SBTi validation, and without disclosed CAPEX allocation is not an institutional-grade target. It is an aspiration. Target Credibility Score: 30/100.

05 — PHYSICAL RISK

BB+	49.5 / 100	47th Percentile	DEVELOPING
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Hazard Exposure Map

HAZARD	SITE	LEVEL	DETAIL	2050 TREND
Water Stress	Raipur	CRITICAL	WRI: 3.8/5 (High) — 62% capacity in stressed zones	Worsening
Heat Stress	Raipur	HIGH	WBGT p95: 29.8°C — at heavy work threshold	Worsening
Cyclone	Vizag	HIGH	Bay of Bengal exposure — 18% assets coastal	Intensifying
Flood	Vizag	MEDIUM	9% assets in 100yr flood zone	Worsening
Sea Level Rise	Vizag	MEDIUM	4.2km coastal — IPCC SLR potential	Accelerating

Water stress is the most pressing physical risk. 62% of manufacturing capacity is in WRI High/Extremely High stress zones. The ZLD system at Raipur provides meaningful mitigation (resilience factor: 0.65 vs 1.0 without). However, ZLD addresses liquid discharge — not raw water procurement risk. No formal water risk assessment or water stewardship strategy was evidenced in any uploaded document.

Heat stress is an emerging operational risk. Raipur's WBGT p95 is 29.8°C — already at the Kjellstrom heavy-work threshold. Under IPCC AR6 3°C projections, this rises to 32.2°C by 2050, projecting 28% heavy-work productivity loss. No heat mitigation CAPEX identified in FY2024 disclosures.

06 — TRANSITION RISK

CCC	28.2 / 100	16th Percentile	LAGGARD
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CRITICAL FINDING: This is BSAL's most material financial risk domain. Carbon pricing alone will consume an estimated **59% of FY2024 EBITDA by 2030** under Disorderly Transition pricing (\$15/tCO2e). At 2040 prices (\$60/tCO2e), the exposure exceeds total current EBITDA.

Carbon Pricing Cost Projection

METRIC	NOW (FY26)	2030	2040	2050
Carbon Price (\$/tCO2e)	\$8–12	\$15–20	\$60–80	\$150–180
BSAL Scope 1 (with traj.)	5,376,000t	4,868,000t	3,782,000t	2,937,000t
Annual Carbon Cost	\$53M	\$73M	\$227M	\$441M
As % of FY24 EBITDA (\$124M)	42.7%	58.9%	183%	356%

Technology lock-in is the most structurally significant transition risk. BSAL's BF-BOF equipment at Raipur was refurbished in 2019, implying remaining asset life to ~2039–2044. No investment in DRI-H2 or EAF technology has been disclosed. Each year of continued BF-BOF investment without a transition plan represents additional stranded asset risk accumulation. Energy intensity at 23.4 GJ/t is 18.2% above the sector median (19.8 GJ/t), driving a **zero score on the energy efficiency sub-component** — the second-largest transition weight (0.25).

07 — SUPPLY CHAIN

D	19.3 / 100	12th Percentile	LAGGARD
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Supply chain risk management is BSAL's weakest domain and near the bottom of its peer cohort. Three structural vulnerabilities define the exposure:

COKING COAL CONCENTRATION

78% of coking coal (1.8 MTPA) sourced from Australia. Queensland flood events (2011, 2022) have previously caused significant global disruption. Single-country concentration at 78% for a CRITICAL input is unmanaged supply chain risk.

SUPPLIER ESG COVERAGE: 12%

Only 17 of 142 Tier 1 suppliers have received any ESG or climate risk assessment. Cohort average: 31%. Best-in-class: 75%+. With Scope 3 Category 1 being the largest single upstream emission source for steel, BSAL cannot credibly claim to manage its full carbon exposure.

NO FORMAL SUPPLIER PROGRAMME

No supplier ESG questionnaire programme is evidenced in any uploaded document. No supplier BCDR requirements. No climate-related supplier contractual clauses identified.

COMPONENT	SCORE	WEIGHT	CONTRIBUTION
Supplier ESG audit coverage (12%)	10.0	0.30	3.0
Scope 3 upstream data quality	15.0	0.25	3.75
Critical material concentration	30.0	0.20	6.0
Supplier questionnaire programme	20.0	0.15	3.0
Supply chain BCDR planning	35.0	0.10	3.5

08 — CLIMATE RISK SIMULATION

Methodology: NGFS 2023 scenarios × IPCC AR6 physical projections. Uncertainty band: ±35% at 80% confidence. Base year: FY2024 revenue \$620M, EBITDA \$124M.

RISK TYPE	SCENARIO	2030	2040	2050
Physical Risk	Orderly NZ2050	\$11.1M	\$24.1M	\$41.6M
Physical Risk	Disorderly ~3°C	\$19.7M	\$44.7M	\$92.4M
Physical Risk	Hot House ~4°C	\$28.1M	\$68.5M	\$151.8M
Transition Risk	Orderly NZ2050	\$82.7M	\$193.2M	\$320.5M
Transition Risk	Disorderly ~3°C	\$128.5M	\$406.4M	\$818.9M
Transition Risk	Hot House ~4°C	\$31.4M	\$45.9M	\$71.9M
COMBINED — % Revenue	Orderly NZ2050	15.1%	35.1%	58.4%

RISK TYPE	SCENARIO	2030	2040	2050
COMBINED — % Revenue	Disorderly ~3°C	23.9%	72.8%	147.0% ■
COMBINED — % Revenue	Hot House ~4°C	9.6%	18.5%	36.1%

KEY INSIGHT: The most financially dangerous scenario for BSAL is Disorderly Transition — not Hot House. Under Hot House (~4°C), carbon prices remain low (governments fail to act) making transition costs modest. Under Disorderly, prices rise faster than BSAL can adapt while physical risk simultaneously intensifies. The 147% revenue exposure by 2050 in Disorderly represents an existential scenario without material decarbonization investment beginning by 2028.

09 — PEER BENCHMARK ANALYSIS

Cohort: Steel — India — Integrated Producer — Large Cap (n=11) | Period: FY2024

DOMAIN	BSAL	MEDIAN	P75	BEST-IN-CLASS	PCT	VS MEDIAN
CIS Overall	40.3	47.2	58.4	78.3	28th	-6.9
Governance	59.8	55.0	68.0	91.0	55th	+4.8
Physical Risk	49.5	52.0	65.0	82.0	47th	-2.5
Transition Risk	28.2	43.0	56.0	79.0	16th	-14.8
Carbon Management	33.0	58.0	72.0	95.0	18th	-25.0
Supply Chain	19.3	35.0	48.0	79.0	12th	-15.7
Compliance	50.2	52.0	65.0	88.0	48th	-1.8
Adaptation	46.9	45.0	58.0	84.0	52th	+1.9
Disclosure Quality	74.4	60.0	72.0	93.0	78th	+14.4
Resilience	13.8	33.0	48.0	79.0	8th	-19.2

10 — PRIORITY ACTION PLAN

TIER 1 — CRITICAL (within 6 months)

01	Conduct formal climate scenario analysis	Directly resolved by BSAL CIS impact: +2.0 Hot House pattern
02	Submit SBTi Commitment Letter	Target credibility: 100% commitment at 2025 (minimum -42%)
03	Commission third-party ESG data assurance (ISAE 3000)	Verification Status: 100% institutional investors FY2025 Scope 1 & 2

TIER 2 — HIGH PRIORITY (within 12 months)

04	Publish standalone carbon inventory (ISO 14064)	Data quality score: 85% gap. Commissioned by 2025
05	Launch formal Supplier ESG Audit Programme	Supply Chain score: 70% 20 suppliers by 2025
06	Commission physical risk assessment — both sites	Physical Risk score: 60% climate risk assessment by 2025

07

Expand executive climate KPIs across full C-suite

Executive acco
rate, supplier a

11 — RATING RATIONALE & INSTITUTIONAL GUIDANCE

Rating: B (40.3) | Outlook: STABLE / POSITIVE POTENTIAL

POSITIVE FACTORS	NEGATIVE FACTORS
+ Consistent BRSR Core disclosure — evidence-verified + Board ESG committee with documented mandate (FY2022) + Paris-aligned Scope 1 intensity trajectory: -2.94%/yr + ZLD system at Raipur — verified CPCB certificate + PAT Scheme Cycle 5 target achieved + Disclosure Quality at 78th peer percentile + HIGH confidence (78) — current, near-complete data	– Scope 1 intensity above sector median → zero performance score – No scenario analysis of any kind conducted or disclosed – Net-Zero 2050 claim with no roadmap or CAPEX plan – Supply chain near-bottom of cohort (12% coverage) – Energy intensity 18.2% above sector median – Climate Resilience at bottom decile (8th percentile) – No third-party ESG data assurance

Upgrade Triggers (to BB or BB+)

- SBTi commitment submitted (achievable in <3 months)
- Third-party ISAE 3000 assurance on FY2025 sustainability data
- Scope 1 intensity falls below 1.85 tCO2e/t (on trajectory: FY2027)
- Formal climate scenario analysis published (TCFD-aligned)

Institutional Investor Guidance

USE CASE	SUITABILITY	RATIONALE
ESG / Climate equity fund	CONDITIONAL HOLD	Monitor SBTi submission. Below Paris-aligned threshold for most mandates.
Investment grade bond / NCD	REQUIRES IMPROVEMENT	Commission ESG assurance before credit mandate. Confidence HIGH but not VERY HIGH.
Green / sustainability-linked bond	NOT ELIGIBLE (current)	No SBTi target. No verified KPI framework. Eligible after SBTi + scenario analysis.
Infrastructure / project finance	PROCEED WITH CONDITIONS	Physical risk assessment required as loan condition. Carbon pricing stress-test in project model.
Pension fund (Paris-aligned mandate)	BELOW THRESHOLD	B rating (28th pct) below typical PA mandate minimums (BB+ / A). Reassess in 12 months.

CLIMATE INTELLIGENCE SCORE	CONFIDENCE	OUTLOOK	ASSESSMENT ID	ENGINE
B (40.3)	HIGH (78)	STABLE / POS.	CX-2026-BSAL-001	v1.0 June 2026